

Sara Alhetela

AI ENGINEER & DATA SCIENTIST

AI engineer and data scientist with a bachelor's in Artificial Intelligence and peer-reviewed research in computer vision and deep learning. Builds and deploys end-to-end machine learning systems: a 18-classifier Python pipeline in production at an energy company and a deep reinforcement learning trading agent trained on 150k+ steps of market data. Skilled across the full model development lifecycle, from experiment tracking and hyperparameter tuning to model deployment and iteration.

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EXPERIENCE

Data Scientist

Feb 2024 - Jul 2024

Fathom.io

- 22% accuracy gain.** Fine-tuned a general-purpose Hugging Face model for task-specific performance, then introduced structured LLM evaluation rubrics and iterative prompt engineering to reduce error rates across key response categories within a four-person Scrum team.
- 35% faster training.** Restructured job scheduling and GPU cluster resource allocation, removing idle compute bottlenecks that slowed weekly model release cycles.
- Five critical bugs resolved.** Applied root-cause analysis to eliminate recurring platform failures, contributing to a 15% uptime improvement over the final two months of the contract.

AI/ML Intern

Jul 2023 - Sep 2023

TAQA - Industrialization & Energy Services Company

- Days to minutes.** Architected and deployed a 3-stage end-to-end ML pipeline in Python predicting cement additive categories, product variants, and concentrations from well parameters — replacing a manual expert process that blocked every pre-construction cycle.
- 99.4% classification accuracy.** Trained 18 ensemble classifiers (ExtraTrees, XGBoost, CatBoost, RandomForest) with per-target hyperparameter optimisation and custom resampling (SMOTE, ADASYN, TomekLinks) to handle class imbalance in real-world oil well data.
- Lab tests eliminated.** Delivered a Flask web interface presenting model outputs to field engineers, removing physical pre-construction testing rounds and cutting turnaround time before drilling began.

AI Workshop Lead

Aug 2021 - Jul 2022

Google Developer Student Club

- Contributed 19 volunteer hours across presenting workshops and club organization activities, covering neural networks, computer vision, and model training in Python. Participants built working models by the end of the sessions with no prior ML background.

PROJECTS

Oil Well Cement Mix Predictor

scikit-learn · XGBoost · CatBoost · Flask

End-to-end 3-stage ML pipeline deployed at TAQA. 187 ensemble classifiers, 99.4% accuracy, per-target hyperparameter tuning, SMOTE/ADASYN resampling, experiment tracking with MLflow, and a Flask web interface used directly by field engineers in pre-construction planning.

Dueling Conv1D DQN Stock Trading Agent

PyTorch · Deep Reinforcement Learning · DQN

Deep reinforcement learning system for algorithmic trading — Dueling Conv1D Q-Network with Double DQN and N-step returns, trained 150k steps on AAPL OHLC data via CUDA. Achieved +18.15% cumulative return on held-out test data vs +3.85% buy-and-hold (+14.30% alpha), 70% trade win rate. Modular architecture across environment simulation, model, training loop, and evaluation with buy-and-hold benchmarking.

github.com/saraalhetela/dueling-dqn-stock-trader

Recyclable Products Classifier

Deep Learning · Computer Vision · TensorFlow

CNN-based image classification model for waste-stream data. Published in Sustainability (2023), vol. 15, no. 14. Peer-reviewed research covering CNN architecture selection and model evaluation.

PUBLICATION

Deep Learning Approach to Recyclable Products Classification: Towards Sustainable Waste Management

Sustainability, vol. 15, no. 14, 2023, p. 11138 · Alhetela, Sara, et al.

EDUCATION

B.Sc. Artificial Intelligence

Imam Abdulrahman Bin Faisal University

Dammam · Aug 2023

GPA 4.5 / 5.0 · First-Class Honours

SKILLS

ML / DL

PyTorch

TensorFlow

Keras

scikit-learn

XGBoost

CatBoost

LLMS & NLP

Hugging Face

Transformers

LangChain

RAG

Prompt Engineering

Fine-tuning

NLP

MLOPS

MLflow

Docker

Kubernetes

CI/CD

FastAPI

Model Deployment

DEEP RL

Deep Reinforcement Learning

DQN

Reward Shaping

CUDA

COMPUTER VISION

CNNs

Image Classification

DATA & CLOUD

Python

SQL

NumPy

Pandas

BigQuery ML

GCP

LANGUAGES

Arabic

Native

English

Advanced

French

Intermediate

CERTIFICATIONS

Google Cloud Skills Boost

- TensorFlow on GCP 2021
- ML Models with BigQuery 2021
- Integrate with ML APIs 2021
- Machine Learning APIs 2021
- Data Engineering 2021

Udemy

- Deep RL using Python 2025

Kaggle

- Feature Engineering 2023
- Intro to Machine Learning 2023